

Kellen Sun

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EDUCATION

University of Waterloo

BCS, Honours Computer Science with Co-Op

Awards: National Math Scholarship, COMC (Top 1%), CCC Senior Honour Roll (Top 5%)

Sept 2023 – Apr 2028

GPA: **3.90/4.0**

EXPERIENCE

Tesla, Palo Alto, California

Jan 2027 – Apr 2027

(Incoming) Software Engineer Intern, AI Compiler | **C++**, **PyTorch**, **MLIR**

- Write, debug and maintain robust software for Tesla AI inference across FSD, Optimus, and Data Center use-cases

Aven Financial, Campbell, California

May 2026 – Aug 2026

Software Engineering Intern | **TypeScript**, **Python**

- Independently shipped a travel portal from database schemas to production, improving search times **7x** by optimizing queries

EnCharge AI, Santa Clara, California

Jan 2026 – Apr 2026

ML Compilers Engineering Intern | **Python**, **C++**, **MLIR**

- Engineered a multi-axis partitioning scheme to support dynamic shapes; spatial splitting to drive a **2.4x** custom analog hardware utilization and temporal splitting to enable execution of larger models exceeding hardware limits
- Enabled support for the Whisper models' architecture by refactoring conv1ds into conv2ds for hardware-accelerated inference while streamlining global tensor shapes, thus eliminating **50%** of costly reshape operations
- Optimized binary sizing estimates, leading to increased L3 memory throughput and a **3%** inference speed improvement
- Architected group convolution decomposition into hardware aligned full convolutions and qint16 into analog efficient qint8s, unlocking native support for new models like ResNet and YoloV8

University of Waterloo, Waterloo, Ontario

Sept 2025 – Dec 2025

Undergraduate Research Assistant | **PyTorch**, **Python**

- Designed a procedural pipeline to synthesize **200,000** valid maze topologies establishing a robust training dataset
- Trained Transformers and Conditional VAEs to generate novel mazes by conditioning latent representations on path length

Ford Motor Company, Ottawa, Ontario

May – Aug 2025 | Sept – Dec 2024

Software/Firmware Developer Intern | **C++**, **Docker**, **Python**, **gtest**, **slash**

- Adapted a POSIX message queue system for Android by implementing user-space integration with Linux kernel's message queue syscalls and file descriptor management while improving unit test coverage for different hardware configurations
- Eliminated thread contentions by rewriting the power management codebase with thread-safe data structures, achieving **100%** line coverage via 15 gtests and developed auto-sleep for the next-gen ECU, saving power across **1000+** units
- Automated testing for an IVI system leveraging new infrastructure which reduced testing time for millions of vehicle by **75%**

PROJECTS

Forge, 🌐; MLX, 🌐 | **Python**, **C++**, **Obj-C**, **Metal Shading Language (MSL)**

- Applied graph optimizations for a custom ML library compiler using kernel fusion leading to a **2x** reduction in memory access
- Crafted custom GPU kernels for matrix multiplication and pointwise operations with various applications in machine learning and image processing resulting in a **13%** speed improvement over PyTorch (mps) on Apple Silicon GPUs
- Enhanced array manipulations and buffer transfers by identifying bottlenecks through profiling then leveraged Apple's unified memory architecture for zero copy transfers, increasing runtime performance by over **30%**
- **MLX** (oss): Fixed silent incorrect NumPy bfloat16 conversions to mx.array and 32-bit overflow issues in the conv3d kernel

AI Chess Transformer, 🌐 | **Python**, **PyTorch**

- Tuned a custom transformer model using a dataset of over **73k** chess games (in PGN format) which generated original 25 move games with over **90%** legal and 80% effective moves by scaling the model to **21.4M** parameters based on this paper 🌐

IMC Prosperity 3, 🌐 | **Python**, **Matplotlib**, **scikit-learn**

- Led a team of 5 to develop an algorithmic trading bot for IMC's Prosperity 3 placing **215th** out of **12,000+** participants
- Optimized models which traded both stable and volatile assets as well as options to both market make and market take depending on strategies such as mean reversion, momentum, statistical arbitrage and delta hedging implied volatility

SKILLS

- **Languages:** C++, C, Python, Metal, TypeScript, Objective-C, CUDA, Rust
- **Libraries and Tools:** PyTorch, NumPy, Git, Bash, Linux/UNIX, MLIR, scikit-learn, pytest, Node.js